EMIRATES GROUP HR — POLICY ASSURANCE CHALLENGE

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Continuous HR policy assurance. **A census, not a sample** — every transaction checked against every written rule, and a named human on every exception.

Repo `github.com/BROCKUGANDA/mizan`

Demo · 2:40 `youtu.be/QE2QcfPyz8Y`

Synthetic data only · MIT licensed · no write path to any HR system

THE PROBLEM

Sampling audits miss what matters

1 in 20

A 5% sample finds roughly one in twenty problems. The other nineteen approve pay, move grades and exit people anyway.

After the fact

Quarterly review reports on a period that already closed. The overpayment is recovered, if at all, months later.

No audit trail

Spreadsheets of findings cannot show who decided what, on which evidence, under whose authority.

MIZAN checks **every** transaction against **every** written clause on every run — in seconds — and turns each failure into an exception that explains itself and waits for a person.

THE APPROACH

A census you can reconcile

1,061 transactions × 15 rule clauses = 15,915 checks → 0.3 s

Applicable + not-applicable = total. Evaluated + not-evaluable = applicable. The arithmetic has to close, and the run asserts it does.

49
findings on the training pack (5 Critical, 32 High, 12 Medium)

13
data-quality exceptions routed to stewardship, not to a payroll owner

22
not evaluable — declared with a DQ code, never a silent pass

0
exceptions closed without a named human decision

Watch the census and queue in 40 seconds · youtu.be/QE2QcfPyz8Y

DETECTION ACCURACY — 30%

Planted violations found, near-misses left alone

CHECK	RESULT
Worked examples in `finding_format.csv`	**5 / 5 reproduce exactly** on every scored column, before the file is allowed to download
Independent SQL oracle (sql.js, written from the policy text)	**100% agreement** with the engine on the full pack
Mutation harness (labelled planted + near-miss variants)	**100% positive recall · 0 near-miss false positives**
Format mutations · metamorphic relations	**9 / 9** rejected · **7 / 7** held
Determinism	Same bytes every time: `ecefa394…` from the CLI, the server and the seed

On the hidden file: `bun run mizan hidden-test <file>` writes `findings.csv` in the required format plus a run manifest — and exits non-zero unless its own census, format and determinism checks pass. A submission cannot be produced from a run that failed itself.

EXPLANATION AND ROUTING — 25%

Every flag argues for itself

F-SSTRR8EN5JC3 Critical New R09 · C1

WX-1003 · Compensation & Payroll Operations

WHAT BROKE

Employee E04567 had an exit date of 2026-02-28, but transaction WX-1003 changed pay by AED 8,200.00 with an effective date of 2026-03-10.

THE RULE (VERBATIM)

> No pay or allowance change may be processed for an employee who is not active on the effective date.

EVIDENCE

amount_fils	AED 8,200.00	transaction
exit_date	**exit 2026-02-28**	employee_master
status at effective date	**ACTIVE → EXITED**	employee_master

WHY IT MATTERS · NEXT ACTION · ROUTING · SEVERITY RATIONALE

Overpayments to leavers are hard to recover → hold payment, confirm status, recover under policy → routed by routing-1.2.0 → CRITICAL because pay eligibility is Critical by policy.

- ✓ **Eight parts** on every finding: headline, what broke, verbatim rule, evidence, why it matters, next action, owner and severity, caveats
- ✓ Every evidence value names **the record it came from**; breaching values are highlighted
- ✓ Severity is a base plus at most two stated modifier steps — the rationale is shown, not implied
- ✓ Routing resolves owner by override → routing table → family overlay, then a **conflict-of-interest guard** refuses to hand it to a party
- ✓ English and Arabic, **RTL first-class**, with SLA due dates from the versioned UAE calendar

Drawer walkthrough at 0:35 · youtu.be/QE2QcfPyz8Y

The engine decides. A person closes.

Model optional, fact-locked

The gateway only rewrites wording. Every number, date and ID must come from the evidence; accusatory phrasing is rejected; any failure falls back to the approved template. `ai.mode=none` is the default and the demo runs that way.

Human decision on every exception

Nothing auto-closes. Dismissing a High or Critical finding requires a **second analyst's countersign** — self-countersigning is refused by the API, as is deciding a case you requested, approved or benefit from.

Audit trail that can be verified

Append-only, hash-chained: every run, decision, countersign, rule change and export. Tampering fails verification and drives the system into safe halt. `mizan audit verify` reproduces the head hash offline.

Data handling

Synthetic data only; a PII scanner blocks real identifiers at upload. Uploads are size-, line- and row-limited with MIME sniffing. Exports escape formula injection. **No write credentials exist** for any source system.

Pattern alerts are advisory only — never routed, never exported, never scored against a person. Small demographic cells (<5) are suppressed. Full statement: `docs/RAI_STATEMENT.md`.

DEPLOYABILITY AND REUSE — 15%

Adding a rule is typing a sentence

```
# Rule studio — plain English, no model required
Any allowance change above AED 10,000 must be
approved by someone at grade 13 or above.

→ comparator >    → quantifier any    (ambiguity chips, not guesses)
→ 3 boundary tests generated
→ backtest 1,061 txns in 0.42 s: would flag 3 · 0.3%
→ overlap: R04, Jaccard 1.0  — it is already policy
```

Two-person approval, then a shadow run, before a rule can go active. Every activation is an immutable version with its own hash.

- ✓ Everything that varies is **versioned data**: routing tables, UAE business calendar, grade ladder, synonyms, severity rubric, SLA policy, action library — editable without a deploy
- ✓ A second business unit loads **its own rule pack**; packs export and import as YAML with tests and a canonical hash
- ✓ Effective-dated weekend change (Fri–Sat → Sat–Sun, 2022) and provisional public holidays are handled as data with caveats
- ✓ Bilingual EN/AR with RTL layout and bidirectional isolates — built for Emirates and reusable across government entities
- ✓ Runs fully offline: no CDN, no telemetry, bundled fonts and icons

TEAM AND PILOT READINESS — 10%

Ready to run a pilot next quarter

Weeks 1–2 · Onboarding

Load the unit's rules, calendar and routing. Roles mapped from the directory. Baseline run, no routing.

Weeks 3–6 · Shadow

Findings stored, nothing routes. Target ≥90% measured precision on exceptions a human has decided.

Weeks 7–12 · Assisted

Live routing and SLA. Weekly metrics: decision coverage 100%, noise ≤10%, deadline adherence ≥90%.

Analyst · Team lead · Operations owner · Policy author · Policy approver · Auditor · Compliance lead · Administrator

Eight roles across ten judge accounts, with a server-enforced permission matrix, queue filters and bulk decisions with typed confirmation, SLA clocks with 75 / 100 / 150% escalation, runbooks RB-01…08 and an on-call rota. Operational modes M0–M5 cover degraded AI, read-only and safe halt — rehearsed, not theoretical.

TRY IT

Runnable in four commands

```
git clone https://github.com/BROCKUGANDA/mizan && cd mizan
bun install
bun run seed     # generates the synthetic pack (seed 42) and loads SQLite
bun run build && bun run start   # http://localhost:3000
```

Repository `github.com/BROCKUGANDA/mizan`

Demo video · 2:40 `youtu.be/QE2QcfPyz8Y`

- bun run verify — 12 assurance gates, all green
- 72 tests
- typecheck + lint clean
- production build
- production seams listed honestly in docs/DEVIATIONS.md

Judge accounts switch from the header — no login, because the sandbox holds no real credentials and the role is checked server-side on every write.

THE ASK

Give MIZAN one HR process for a quarter

Pick the noisiest process — allowance changes or exits — and run twelve weeks in shadow against its real transaction history. The measure is not how many findings MIZAN produces; it is **decision coverage at 100%, noise under 10% and deadlines met nine times in ten**, with every one of them traceable to a rule, a record and a named person.

30%
of the score is detection — MIZAN proves it against an independent oracle, not against itself

45%
is explanation, routing and responsible AI — the three things an audit function actually has to defend

0
real employee data touched, and none required to start the pilot in shadow

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youtu.be/QE2QcfPyz8Y